

What Goes On in the Courtroom? Audiovisual Analysis of Ninth Circuit Oral Arguments

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Research Question + Data

- What Goes on in the Courtroom?
 - Do judges convey information through body language and speech?
 - Can this information be extracted at scale by a computer?
- Data
 - ~10,000 videos downloaded from Ninth Circuit's YouTube Channel
 - LexisNexis data with judge names, appointing President, votes, case characteristics



Previous Literature

- Emotional Arousal in audio to predict Supreme Court votes (Dietrich, Enos, and Sen 2019)
- Audio analysis of telephone interviews (Dietrich, Mondak, Williams 2019)
- Measuring polarization with C-SPAN videos (Dietrich 2021)
- Analyzing political slant of images during movements like BLM (Kim and Bas 2023)

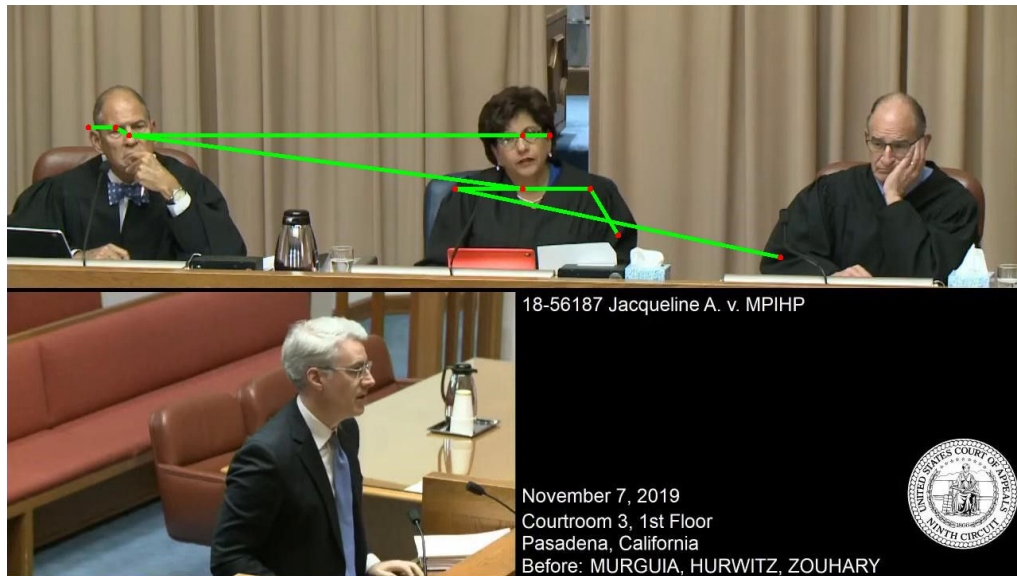
Judge Identification and Facial Emotion Recognition

- CVAT Labeling
 - Ninth Circuit Judges
 - Not labeled:
 - Visiting judges
 - Attorneys
 - Audience members/clerks/staff
- Facial Recognition
 - Two steps
 - Identify human faces and draw bounding boxes
 - Make prediction about who is in the bounding box (Haar Cascades, OpenFace Deep Neural Net, custom Convolutional Neural Net)
- Facial Emotion Recognition
 - One of seven emotions (happy, angry, disgusted, fearful, neutral, sad, surprised)
 - Record max emotion and emotion probability proportions



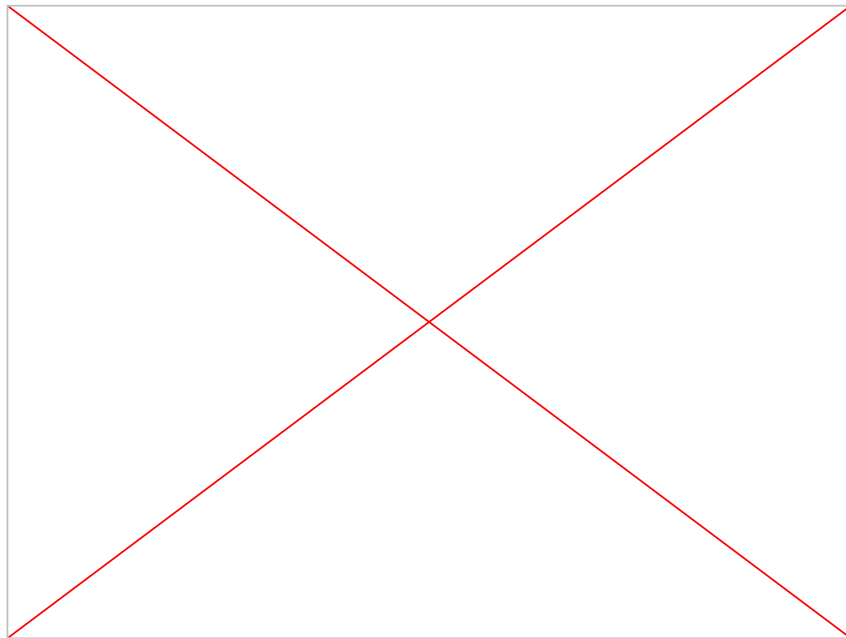
Judge Poses

- OpenPose framework
- Detect features corresponding to human body parts
- Construct features pairs to estimate “poses”
- To Do:
 - Label and classify specific poses
 - Match poses with outcomes like reading notes, looking at attorney/colleagues, etc.



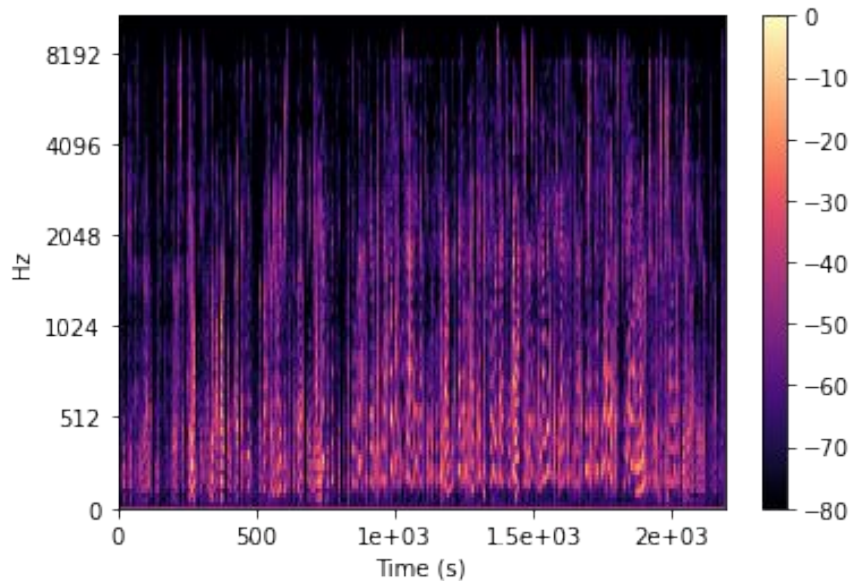
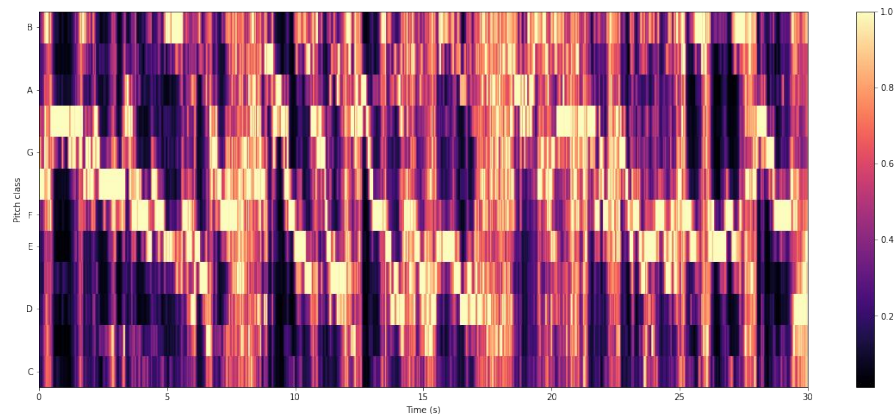
Speaker Diarization and Transcription

- Diarization and Transcription
 - OpenAI Whisper model for automatic diarization and transcription of text
 - Does NOT do speaker recognition
- Pipeline for segmenting by speaker and labeling transcription with our audio classification model

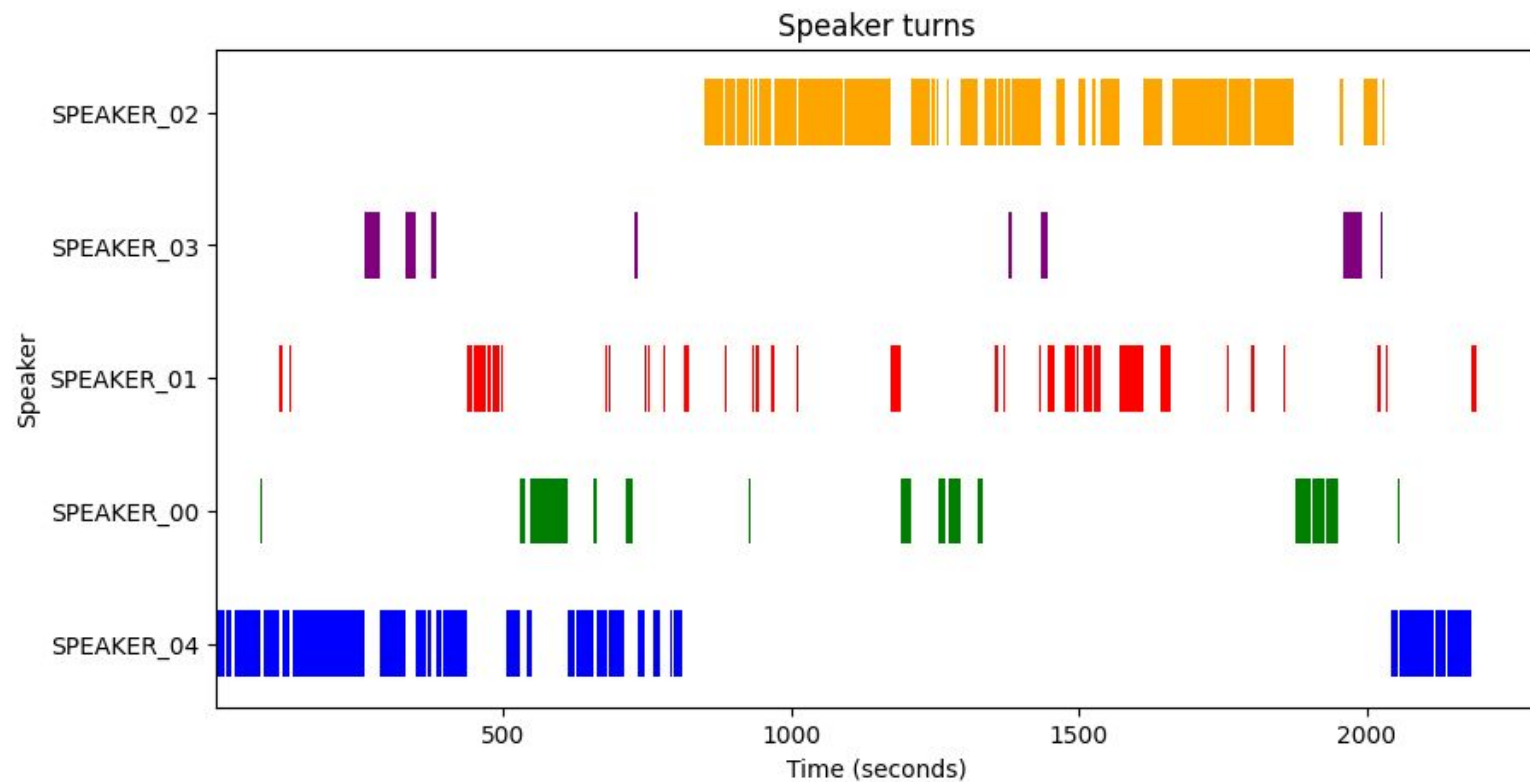


Audio Features

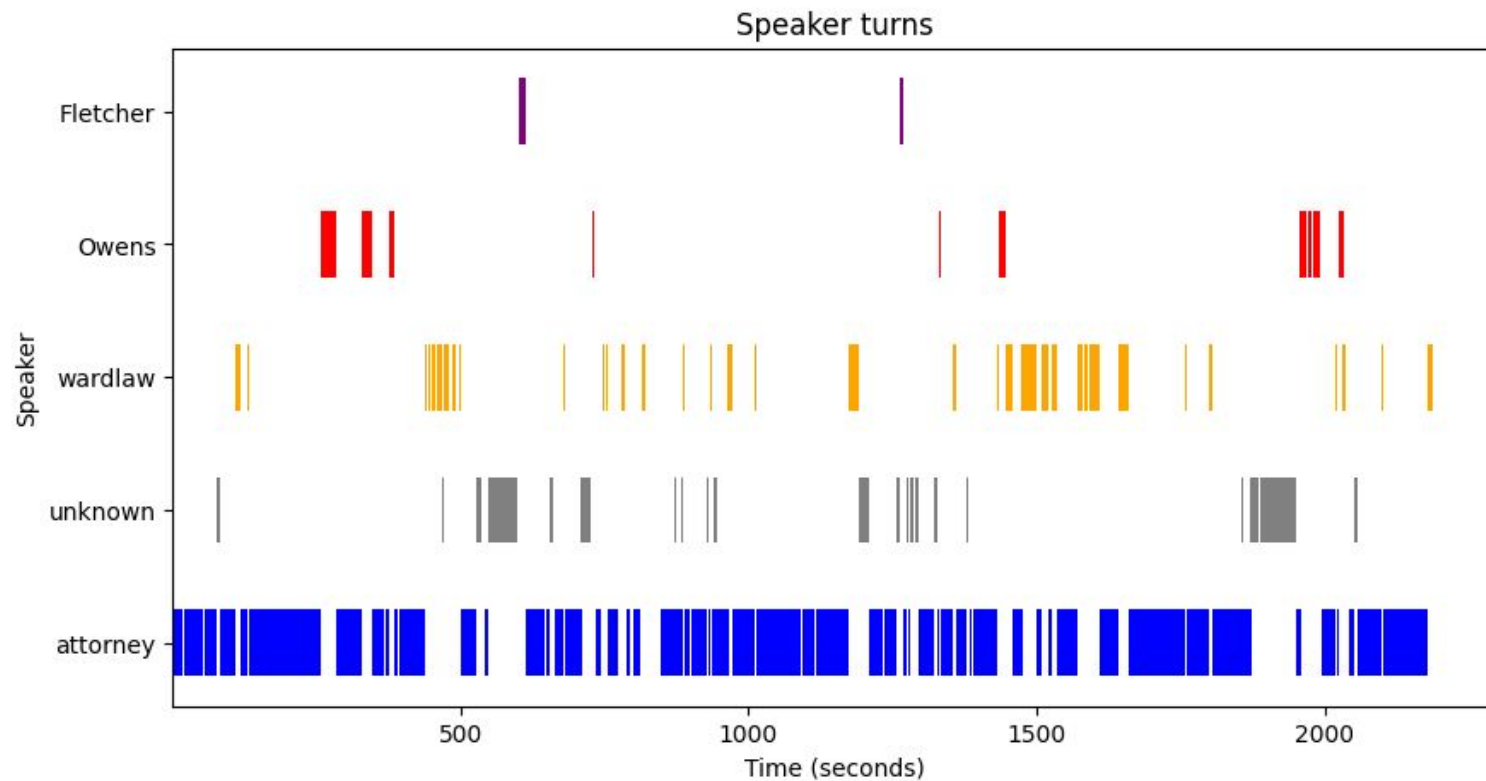
- Pitch, frequency, and amplitude
- Mel-frequency cepstral coefficients (MFCCs)
- Distinguish speakers
- “Emotional Arousal”



Diarization



Speaker Identification



Transcription

attorney: The expert testified for over an hour as to his qualifications. The experts testified that he had been a citizen in the United States since 2000., '

fletcher: Was there some reason why the expert didn't want to provide the A number? Was it hard to find?, '

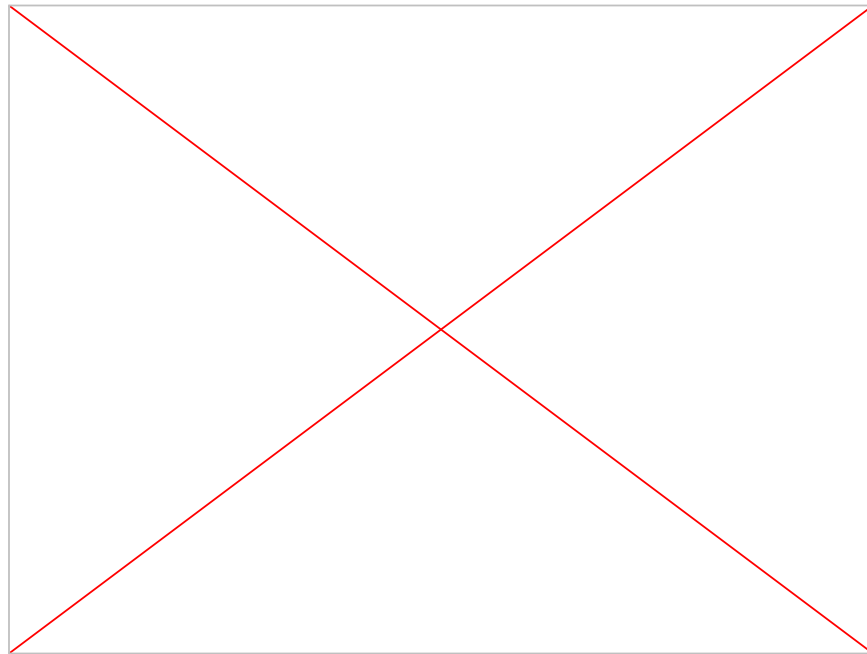
attorney: The declaration of the expert states that he felt that he had been was being abused by the judge and that he did not want to provide any further testimony than he had., '

attorney: His declaration, the most emotional accuse, states that he was being unreasonably challenged by the judge, the judge is rolling her eyes., '

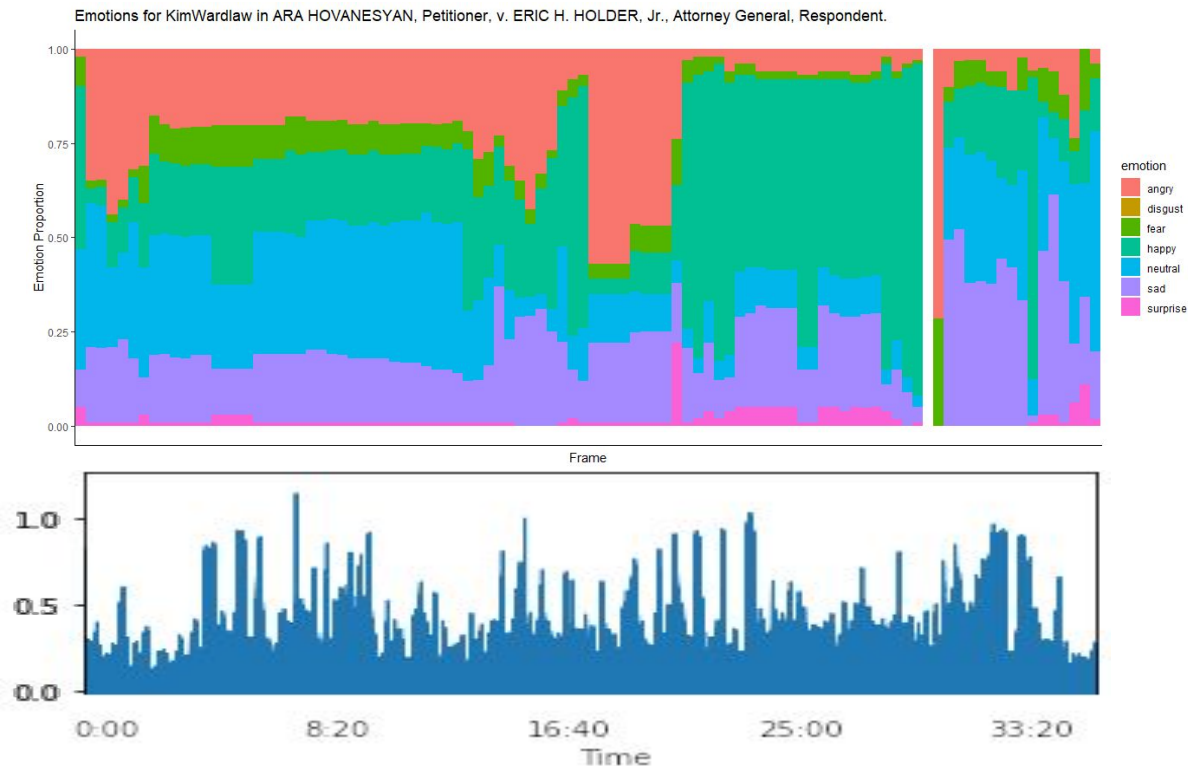
wardlaw: Is it an extended practice to ask for the A number of witnesses and immigration proceedings?, '

attorney: That would be true for a lay witness, Your Honor, but not necessarily for an expert witness. I don't see the relevance of providing an A number for the expert witnesses., '

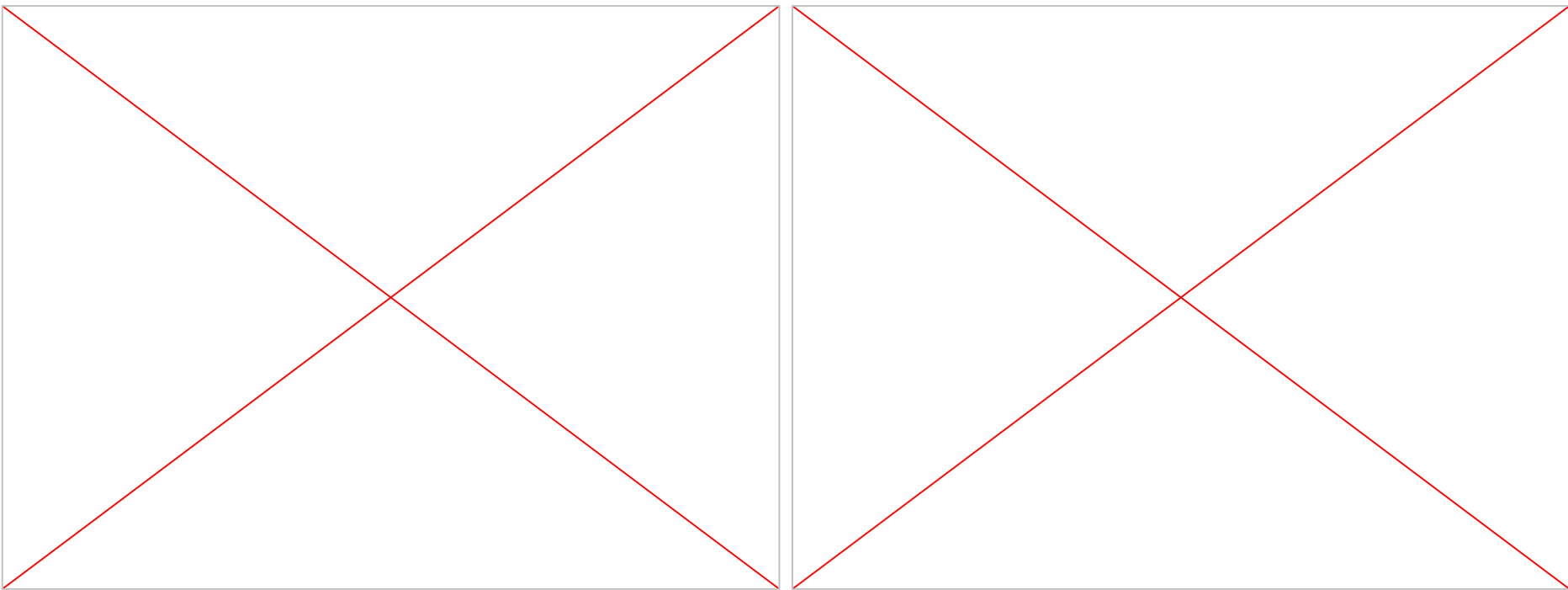
wardlaw: Why are we assuming that the witness has an A number?, '



Tracking a Judge Through a Video



Tracking Judges Through a Video



Generative AI Approach to Courtroom Narratives

Define a prompt that includes the frame number consideration and transcript text

```
prompt = f"""
```

```
    Finally, analyze the following courtroom scene with three judges presiding over a case.
```

```
    Consider the following frame number range {start_frame} to {end_frame} where the attorney is speaking.
```

```
    The attorney said: "{transcript_text}"
```

```
    For each judge, provide the following details and structure the output as follows:
```

```
    - Judge's Position: [left, right, center]
```

```
    - Body Language:
```

```
    - Facial Expression:
```

```
    - Gestures: (hand movements, head nods, etc.)
```

```
    - Gaze: (who or what they are looking at)
```

```
    - Change: (analyze any changes relative to the first two frames)
```

```
    - ResponseToAttorney: (analyze the judge's facial reaction when the attorney is speaking.
```

```
    Make specific references to the content of what the attorney has said and make your best guess about what the judge's reaction is to that content.)
```

```
    Only return your analysis for this final frame, the first two frames were provided as reference solely to populate the Change
```

```
    """
```

Generative AI Approach to Courtroom Narratives

```
{'frame_0': {'Judge1': {'name': 'WARDLAW',  
  'position': 'center',  
  'Body Language': 'Sitting upright, hands resting on the table, slightly turned towards the attorney.',  
  'Facial Expression': 'Curious, with a slight furrowed brow.',  
  'Gestures': 'Minimal gestures, occasionally adjusting glasses or papers.',  
  'Gaze': 'Maintaining eye contact with the attorney, indicating interest in the argument.',  
  'Change': 'The judge's expression has shifted from passive to more engaged, reflecting a readiness to hear the case.',  
  'ResponseToAttorney': 'The slight furrow in the brow may indicate the judge is considering the implications of the attorney's introduction.'}},  
  
'frame_750': {'Judge1': {'name': 'WARDLAW',  
  'position': 'center',  
  'Body Language': 'Sitting upright, hands clasped in front.',  
  'Facial Expression': 'Serious, with a slight frown.',  
  'Gestures': 'Minimal hand movements, focused on the attorney.',  
  'Gaze': 'Alternating between the attorney and notes.',  
  'Change': 'The judge's serious demeanor has intensified, reflecting the gravity of the attorney's statement.',  
  'ResponseToAttorney': 'The judge's frown deepens when the attorney discusses the ultimatum, indicating disapproval or concern regarding the treatment of the expert.'}},  
  
'Judge3': {'name': 'OWENS',  
  'position': 'right',  
  'Body Language': 'Relaxed posture, hands clasped in front.',  
  'Facial Expression': 'Calm, with a slight smile, indicating openness.',  
  'Gestures': 'Occasional nodding, showing understanding or agreement.',  
  'Gaze': 'Alternating between the attorney and the other judges, maintaining awareness of the discussion.',  
  'Change': 'The judge appears more relaxed compared to the earlier frames, possibly indicating comfort with the ongoing dialogue.',  
  'ResponseToAttorney': 'The slight smile and nodding suggest the judge finds the attorney's argument relatable or valid, particularly the reference to actors in political positions.'}},
```

Generative AI Approach to Courtroom Narratives

Judge Wardlaw (Center)

- Plaintiff's Attorney:
 - In the initial part of the plaintiff's argument, Judge Wardlaw is seen with a slightly furrowed brow and serious demeanor, indicating deep contemplation of the attorney's introduction (gpt_example_transcript_...). As the plaintiff's attorney discusses the importance of the expert witness and the "A number" (a key identification number), Wardlaw's seriousness intensifies, suggesting that she is critically assessing the weight of this evidence (gpt_example_transcript_...). At one point, when the attorney introduces an ultimatum regarding the expert witness, Wardlaw's expression deepens into a frown, signaling disapproval or concern over the implications of the expert's treatment (gpt_example_transcript_...).
- Respondent's Attorney:
 - As the respondent's attorney begins their arguments, Wardlaw maintains her serious posture, arms crossed, but her reactions to claims about the witness's visa change show critical evaluation. The furrowed brow indicates skepticism about the expert's qualifications and their impact on the case (gpt_example_transcript_...). Wardlaw's serious demeanor persists when the respondent's attorney discusses the government's handling of expert testimony, reflecting her careful weighing of the procedural context (gpt_example_transcript_...).

Judge W. Fletcher (Left)

- Plaintiff's Attorney:
 - During the plaintiff's argument, Fletcher's posture shifts slightly forward, signaling attentiveness and engagement (gpt_example_transcript_...). His neutral expression suggests he is open to hearing the attorney's points, particularly when the attorney emphasizes the significance of the expert's role. His subtle reactions, such as a slight furrowed brow when discussing the expert's qualifications, indicate Fletcher's consideration of the legal weight behind the claims (gpt_example_transcript_...).
- Respondent's Attorney:
 - In the second half, when the respondent's attorney highlights the expert's credentials and visa status, Fletcher continues to lean forward and displays a hint of concentration (gpt_example_transcript_...). His neutral expression indicates he is processing the respondent's claims, and at times his hand movements suggest thoughtfulness. His facial expressions remain relatively unchanged, but his engagement suggests that he is weighing the relevance of the expert's testimony (gpt_example_transcript_...).

Judge Owens (Right)

- Plaintiff's Attorney:
 - Owens adopts a relaxed posture throughout the plaintiff's arguments, occasionally smiling or nodding, which suggests a degree of receptivity to the points being made (gpt_example_transcript_...). His calm demeanor and subtle nodding indicate agreement or understanding of the arguments related to the expert witness's treatment. However, Owens remains less expressive than the other judges, often alternating between looking at the attorney and glancing at notes or the other judges (gpt_example_transcript_...).
- Respondent's Attorney:
 - During the respondent's argument, Owens continues his relaxed posture but becomes more contemplative. His occasional nodding indicates he is following the respondent's claims regarding the expert's qualifications (gpt_example_transcript_...). At certain points, Owens tilts his head slightly, signaling curiosity or doubt about the relevance of the expert's visa change and qualifications (gpt_example_transcript_...).

Next Steps

- Labeling
- Automatically detecting key events and Higher Level Features
 - Judge making eye contact with colleagues and attorney, or looking at notes?
 - Interrupting attorneys and colleagues?
 - Relative to their own baselines, are judges bored? Sleepy? Engaged?
- Case-level analysis
 - Do judges behave differently in criminal and civil suits?
 - Do Republicans and Democrats behave differently?
 - How predictive are audio/video features of judge votes?
- Causal Inference
 - Does the introduction of video change the way judges and attorneys speak in the courtroom?
- Other future directions?
 - Applications of video + audio + generated text
 - Sentiment analysis
 - Audiovisual topic modeling in combination with text
 - Other areas?